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Prof. Valérie Gouranton (IRISA)
Prof. Jean-Yves Didier (Université d'Évry)

Subject: Application for PhD Position – Navigating through the temporal aspects of the digital twin

Dear Prof. Gouranton and Prof. Didier,

I am applying for the PhD position on "Navigating through the temporal aspects of the digital twin" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

Managing the temporal state of a digital twin is critical for predictive analytics. During my work on the Gas Turbine Digital Twin, I implemented Mamba-SSM sequence models specifically to capture long-term temporal dependencies and enable generative 'What-If' forecasting. I am eager to expand this work by developing structured navigation frameworks that allow users to seamlessly traverse past telemetry, present states, and future predictive timelines within the digital replica.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui